

The Class of Semi Doubly Stochastic Matrices

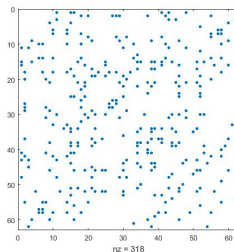
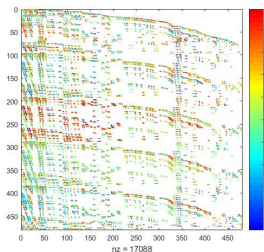
A Novel Scaling Approach for Rectangular Matrices

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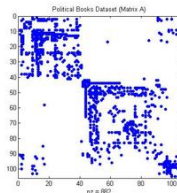
Motivations



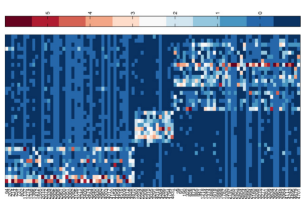
- How do we balance size and structure of nonzeros in a matrix to draw out key features?
- Can we make effective use of spectral information?

Motivations

- Preprocessing for large sparse matrices
- Linear algebra
- Community detection



- Genomics: biclustering on microarray data



Scaling to Constant Row and Column Sums

Given a $m \times n$ matrix $\mathbf{A} \geq 0$, finding diagonal matrices $\mathbf{D}, \mathbf{E}$ s.t. $\mathbf{S} = \mathbf{DAE}$ has constant row and column sums.

Assume that $\mathbf{A}$ is **connected** (bipartite matrix of a connected graph):

$$\mathbf{A} = \begin{matrix} & \begin{matrix} c_1 & c_2 \end{matrix} \\ \begin{matrix} r_1 \\ r_2 \\ r_3 \end{matrix} & \begin{bmatrix} \times & \\ \times & \times \\ & \times \end{bmatrix} \end{matrix} \iff \begin{matrix} r_1 & \diagdown & c_1 \\ & \diagup & \\ r_2 & \diagdown & c_2 \\ & \diagup & \\ r_3 & \diagdown & c_2 \end{matrix}$$

- If $\mathbf{A}$ is square, try to scale to doubly stochastic:

$$\mathbf{S}\mathbf{1} = \mathbf{1} \quad \text{and} \quad \mathbf{S}^T\mathbf{1} = \mathbf{1}.$$

- Otherwise, try to scale to:

$$\mathbf{S}\mathbf{1}_n = \sqrt{\frac{n}{m}}\mathbf{1}_m \quad \text{and} \quad \mathbf{S}^T\mathbf{1}_m = \sqrt{\frac{m}{n}}\mathbf{1}_n.$$

Scaling to (Piecewise) Constant Row and Column Sums

Now assume that $\mathbf{A} = \begin{bmatrix} \mathbf{A}_1 & & \\ & \ddots & \\ & & \mathbf{A}_k \end{bmatrix} :$

- If $\mathbf{A}$ is square, still scale to doubly stochastic: $\mathbf{S} = \mathbf{DAE}$ s.t

$$\mathbf{S}\mathbf{1} = \mathbf{1} \text{ and } \mathbf{S}^T\mathbf{1} = \mathbf{1}.$$

- Otherwise, scale to: $\mathbf{S} = \mathbf{DAE} = \begin{bmatrix} \mathbf{S}_1 & & \\ & \ddots & \\ & & \mathbf{S}_k \end{bmatrix}$ s.t

$$\forall i, \mathbf{S}_i \mathbf{1}_{n_i} = \sqrt{\frac{n_i}{m_i}} \mathbf{1}_{m_i} \text{ and } \mathbf{S}_i^T \mathbf{1}_{m_i} = \sqrt{\frac{m_i}{n_i}} \mathbf{1}_{n_i}.$$

Scaling to (Piecewise) Constant Row and Column Sums

Classic scaling approaches scale to prescribed row and column sums:
Fixing two vectors $\mathbf{r}, \mathbf{c}$, they return diagonal matrices $\mathbf{D}, \mathbf{E}$ s.t.

$$\begin{cases} \mathbf{DAE}\mathbf{1}_m = \mathbf{r} \\ \mathbf{EA}^T\mathbf{D}\mathbf{1}_n = \mathbf{c} \end{cases}$$

✓ If $\mathbf{A}$ is square then $\mathbf{r} = \mathbf{c} = \mathbf{1}_n$.

⚠ Otherwise... it's tricky.

✗ Besides, with a bad choice of $\mathbf{r}, \mathbf{c}$, algorithms fail.

But with $\mathbf{S} = \mathbf{DAE}$ s.t. $\mathbf{S}_i\mathbf{1}_{n_i} = \sqrt{\frac{n_i}{m_i}}\mathbf{1}_{m_i}$ and $\mathbf{S}_i^T\mathbf{1}_{m_i} = \sqrt{\frac{m_i}{n_i}}\mathbf{1}_{n_i}, \forall i$:

$$\mathbf{SS}^T\mathbf{1}_m = \mathbf{1}_m \text{ and } \mathbf{S}^T\mathbf{S}\mathbf{1}_n = \mathbf{1}_n$$

$\implies$ Normal equations of $\mathbf{S}$ are doubly stochastic.

To Summarise: Scaling Rectangular Matrices

Scaling a $m \times n$ matrix $\mathbf{A} \geq 0$:

Classic approach

Given two vectors $\mathbf{r}, \mathbf{c}$, finding diagonal matrices with positive diagonal $\mathbf{D}, \mathbf{E}$ s.t.

$$\begin{cases} \mathbf{DAE}\mathbf{1}_m = \mathbf{r} \\ \mathbf{EA}^T\mathbf{D}\mathbf{1}_n = \mathbf{c} \end{cases}$$

⚠ How to choose a consistent pair $\mathbf{r}, \mathbf{c}$?

Proposed circumvention

Finding $\mathbf{D}, \mathbf{E}$ such that $\mathbf{S} = \mathbf{DAE}$ verifies

$$\begin{cases} \mathbf{SS}^T\mathbf{1}_m = \mathbf{1}_m \\ \mathbf{S}^T\mathbf{S}\mathbf{1}_n = \mathbf{1}_n \end{cases}$$

✓ No need to fix $\mathbf{r}, \mathbf{c}$.

Semi Doubly Stochastic Matrices

Relation with Doubly Stochastic Matrices

Definition

A is called Semi Doubly Stochastic (SDS) iff

$$\mathbf{A}^T \mathbf{A} \mathbf{1}_n = \mathbf{1}_n \text{ and } \mathbf{A} \mathbf{A}^T \mathbf{1}_m = \mathbf{1}_m.$$

- Any doubly stochastic (DS) matrix is SDS:

$$\mathbf{A} \mathbf{1} = \mathbf{1} \text{ and } \mathbf{A}^T \mathbf{1} = \mathbf{1} \implies \mathbf{A} \mathbf{A}^T \mathbf{1} = \mathbf{1} \text{ and } \mathbf{A}^T \mathbf{A} \mathbf{1} = \mathbf{1}.$$

- But the reverse is not true:

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 1/\sqrt{2} \\ 0 & 0 & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \end{bmatrix} \implies \mathbf{A} \mathbf{A}^T = \mathbf{A}^T \mathbf{A} = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\implies$ SDS matrix class contains all the DS matrices, but not only.

Semi Doubly Stochastic Matrices

Characterisation

Theorem

A $m \times n$ connected matrix $\mathbf{A} \geq 0$ is SDS $\iff$

$$\mathbf{A}\mathbf{1}_n = \sqrt{\frac{n}{m}}\mathbf{1}_m \text{ and } \mathbf{A}^T\mathbf{1}_m = \sqrt{\frac{m}{n}}\mathbf{1}_n.$$

Proof: ($\Rightarrow$) If $\mathbf{A}$ SDS, denote $\mathbf{f} = \mathbf{A}\mathbf{1}_n$. Then

$$\mathbf{A}^T\mathbf{f} = \mathbf{1}_n \text{ and so } \mathbf{A}\mathbf{A}^T\mathbf{f} = \mathbf{f}.$$

$\mathbf{A}\mathbf{A}^T$ bi-irreducible ($\mathbf{A}$ is connected) and stochastic, the Perron-Frobenius theorem implies that $\mathbf{f} = \alpha\mathbf{1}_m$ with $\alpha > 0$.

$$\text{Finally, } n = \mathbf{1}_n^T\mathbf{1}_n = (\mathbf{A}^T\mathbf{f})^T(\mathbf{A}^T\mathbf{f}) = \mathbf{f}^T\mathbf{f} = m\alpha^2 \implies \alpha = \sqrt{n/m}.$$

Same reasoning with $\mathbf{g} = \mathbf{A}^T\mathbf{1}_m$.

($\Leftarrow$) Direct computation.

Semi Doubly Stochastic Matrices

Characterisation

Theorem

A $m \times n$ matrix $\mathbf{A} \geq 0$ is SDS iff it is the direct sum of $m_i \times n_i$ connected matrices $\mathbf{A}_i$ such that

$$\mathbf{A}_i \mathbf{1}_{n_i} = \sqrt{n_i/m_i} \mathbf{1}_{m_i} \quad \text{and} \quad \mathbf{A}_i^T \mathbf{1}_{m_i} = \sqrt{m_i/n_i} \mathbf{1}_{n_i}$$

Proof: Straightforward since any $\mathbf{A}$ can be permuted into

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_1 & & \\ & \ddots & \\ & & \mathbf{A}_k \end{bmatrix},$$

each $\mathbf{A}_i$ being connected.

Semi Doubly Stochastic Matrices

Additional Properties

Property

A square matrix $\mathbf{A} \geq 0$ is DS iff it is SDS and has support.

Proof: ($\Rightarrow$) Straightforward. ($\Leftarrow$) The connected blocks of the square matrix $\mathbf{A}$ must be square. Otherwise, $\mathbf{A}$ has no support:

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_1 & & \\ & \ddots & \\ & & \end{bmatrix} = \left[\begin{array}{ccc|cc} \otimes & \times & \times & & \dots \\ \times & \otimes & \times & & \dots \\ \hline & & & \otimes & \times \\ & & & \times & \otimes \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{array} \right]$$

If $m_1 < n_1$, no diag of $\mathbf{A}$ covers all $\mathbf{A}_1$ columns. Thus $\mathbf{A}$ has no support. Same reasoning if $m_1 > n_1$.

Thus, all connected blocks of the SDS matrix are square and thus DS:

$$\mathbf{A}_i \mathbf{1}_{n_i} = \sqrt{n_i/n_i} \mathbf{1}_{n_i} = \mathbf{1}_{n_i} \quad \text{and} \quad \mathbf{A}_i^T \mathbf{1}_{n_i} = \sqrt{n_i/n_i} \mathbf{1}_{n_i} = \mathbf{1}_{n_i},$$

which enforces $\mathbf{A}$ to be DS.

Semi Doubly Stochastic Matrices

Additional Properties

Property

A connected matrix $\mathbf{A}$ is SDS implies: (1) $\mathbf{A}$ has the Strong Hall Property and (2) every nonzero of $\mathbf{A}$ lies on a full diagonal.

Proof: $\mathbf{A}$ connected $\implies ((1) \iff (2))$ (Th3.3 of *Brualdi&Shader'94*).
A SDS $m \times n$ (with $m \geq n$) connected matrix $\mathbf{A}$ verifies:

$$\begin{aligned} \forall I \times J \subset \{1, \dots, m\} \times \{1, \dots, n\}, \quad \mathbf{A}(I, J) = 0 &\implies |J|\sqrt{m/n} < |\bar{I}|\sqrt{n/m} \\ &\iff \underbrace{\frac{m}{n}|J|}_{\geq 1} + |I| < m \\ &\implies |J| + |I| < m \end{aligned}$$

First line is a direct consequence of (Th3.5 of *Menon&Schneider'69*).
Last line defines a Strong Hall connected matrix (*Brualdi&Shader'94*).

Semi Doubly Stochastic Matrices

Additional Properties

Counter-Example

Strong Hall Property does not imply SDS:

$$\mathbf{A} = \begin{matrix} & & \sqrt{2} & \sqrt{2} \\ \begin{matrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 1/\sqrt{2} \\ 1/\sqrt{2} \end{matrix} & \begin{bmatrix} \times & \times \\ \times & \times \\ & \times \\ & \times \end{bmatrix} \end{matrix} \Rightarrow \begin{bmatrix} \times & \mathbf{0} \\ \times & \mathbf{0} \\ & 1/\sqrt{2} \\ & 1/\sqrt{2} \end{bmatrix}$$

$\Rightarrow$ Given a connected $m \times n$ matrix $\mathbf{A}$ ($m \geq n$), a SDS matrix with same pattern as $\mathbf{A}$ exists iff

$$\forall I \times J \subset \{1, \dots, m\} \times \{1, \dots, n\}, \quad \left(\mathbf{A}(\bar{I}, J) = 0 \Rightarrow \frac{|J|}{|I|} < \frac{n}{m} \right),$$

(direct consequence of Th3.5 of *Menon&Schneider'69*).

Semi Scalability

Definition

A matrix $\mathbf{A} \geq 0$ is called **Semi Scalable** (SS) if it is diagonally equivalent to a SDS matrix. That is there exist two diagonal matrices $\mathbf{D}, \mathbf{E}$ with positive diagonal s.t. $\mathbf{S} = \mathbf{DAE}$ is SDS.

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Examples:

$$\begin{bmatrix} \frac{1}{\sqrt[4]{2}} & & \\ & \frac{1}{\sqrt[4]{2}} & \\ & & \frac{1}{\sqrt[4]{2}} \end{bmatrix} \times \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \times \begin{bmatrix} \frac{1}{\sqrt[4]{2}} & & \\ & \frac{1}{\sqrt[4]{2}} & \\ & & \frac{1}{\sqrt[4]{2}} \end{bmatrix} = \begin{bmatrix} 0 & 0 & \frac{1}{\sqrt{2}} \\ 0 & 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{bmatrix}$$

This matrix is SS.

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Examples:

$$\begin{bmatrix} \alpha & \\ & \beta \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \delta & \\ & \gamma \end{bmatrix} = \begin{bmatrix} \alpha\gamma & \alpha\delta \\ 0 & \beta\delta \end{bmatrix}$$

This matrix cannot be SS

$\Rightarrow$ A square matrix with support but no total support: we knew it.

Semi Scalability

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Property (from Th2 of *Menon'66*)

A matrix $\mathbf{A} \geq 0$ is SS iff it exists a SDS matrix with the same pattern as $\mathbf{A}$.

Scaling Matrices to SDS Form

An Algorithm

Input: A $m \times n$ matrix $\mathbf{A} \geq 0$, with no zero row or column

Output: Two diagonal matrices $\mathbf{D} \in \mathbb{R}^{m \times m}$ and $\mathbf{E} \in \mathbb{R}^{n \times n}$, and $\mathbf{S} \in \mathbb{R}^{m \times n}$ such that $\mathbf{S} = \mathbf{DAE}$

- 1: $\mathbf{A}^{(0)} \leftarrow \mathbf{A}$
- 2: $\mathbf{d}^{(0)} \leftarrow \mathbf{1}_m$
- 3: $\mathbf{e}^{(0)} \leftarrow \mathbf{1}_n$
- 4: **for** $k = 0, 1, 2, \dots$ **until** convergence **do**
- 5: $\mathbf{r} \leftarrow \mathbf{A}^{(k)} \mathbf{A}^{(k)T} \mathbf{1}_m$
- 6: $\mathbf{c} \leftarrow \mathbf{A}^{(k)T} \mathcal{D}(\mathbf{r})^{-1} \mathbf{A}^{(k)} \mathbf{1}_n$
- 7: $\mathbf{A}^{(k+1)} \leftarrow \mathcal{D}(\sqrt{\mathbf{r}})^{-1} \mathbf{A}^{(k)} \mathcal{D}(\sqrt{\mathbf{c}})^{-1}$
- 8: $\mathbf{d}^{(k+1)} \leftarrow \mathcal{D}(\sqrt{\mathbf{r}})^{-1} \mathbf{d}^{(k)}$
- 9: $\mathbf{e}^{(k+1)} \leftarrow \mathcal{D}(\sqrt{\mathbf{c}})^{-1} \mathbf{e}^{(k)}$

Set $\mathbf{D} = \mathcal{D}(\mathbf{d}^{(k+1)})$, $\mathbf{E} = \mathcal{D}(\mathbf{e}^{(k+1)})$, and $\mathbf{S} = \mathbf{A}^{(k+1)}$

Scaling Matrices to SDS Form

Convergence of the Algorithm

Theorem

Let $\mathbf{A} \geq 0$, and $\mathbf{B}$ a SDS matrix with the same pattern ($\mathcal{P}(\mathbf{A}) = \mathcal{P}(\mathbf{B})$). Then, the scaling algorithm produces a sequence of iterates $(\mathbf{A}^{(k)})_{k \geq 0}$ ($\mathbf{A}^{(0)} = \mathbf{A}$) that converges to a semi-doubly stochastic limit $\mathbf{Q}$. The scaling factors $(\mathbf{d}^{(k)}, \mathbf{e}^{(k)})_{k \geq 0}$ also have a limit, $(\mathbf{d}, \mathbf{e})$ say, and

$$\mathbf{Q} = \mathcal{D}(\mathbf{d})\mathbf{A}\mathcal{D}(\mathbf{e}).$$

Corollary

A matrix $\mathbf{A} \geq 0$ is SS $\iff$ the algorithm converges (e.g. $(\mathbf{d}^{(k)}, \mathbf{e}^{(k)})_k$ have a limit).

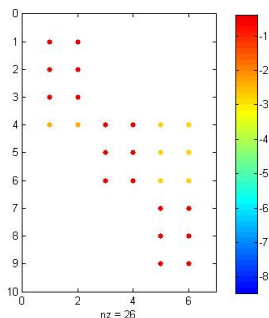
Keypoint: The algorithm is Diagonal Product Increasing. The rest is from *Pretzel'80* adapted to SDS.

When the matrix is not SS...

The proof of the convergence of the algorithm is not easy to extend to the non SS case, because the targets are not prescribed.

We observe that:

- The sequence of matrices $(\mathbf{A}^{(k)})_k$ still converges, while the scaling factors diverge. Convergence can be terribly slow though.
- Some elements vanish in $\lim_{k \rightarrow \infty} \mathbf{A}^{(k)}$, creating disconnected blocks.



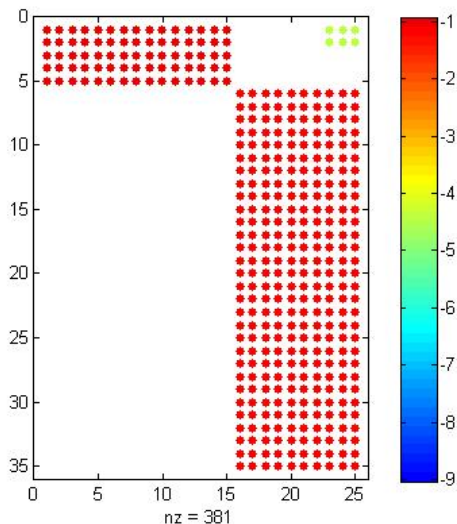
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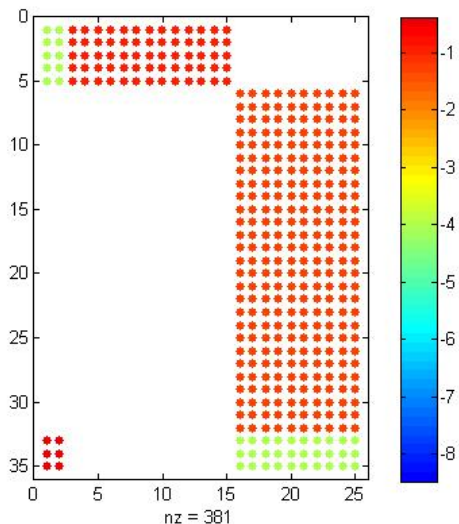
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- The sequence of matrices $(\mathbf{A}^{(k)})_k$ still converges, while the scaling factors diverge. Convergence can be terribly slow though.
- Some elements vanish in $\lim_{k \rightarrow \infty} \mathbf{A}^{(k)}$, creating disconnected blocks.
- There always exists a SS sub-matrix within $\mathbf{A}$.
- But the algorithm does not converge to the one with the maximum number on non-zeros.
- For now, we cannot predict which elements will vanish.

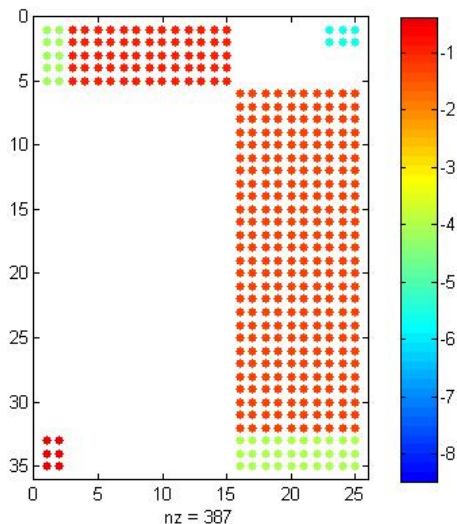
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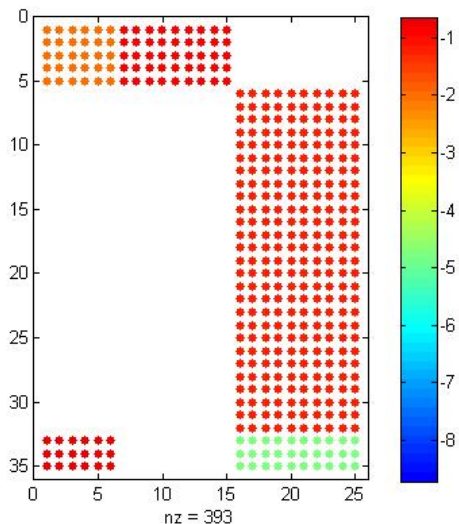
When the matrix is not SS...



When the matrix is not SS...



When the matrix is not SS...



Thank you !

Questions ?